Big Bang Theory (Georges Lemaître)
Sunday Expositions Series(#2)

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March 4, 2026

THE PRIMEVAL ATOM (THE SUPERBASIC)

This chapter establishes the foundational paradigm shift from a static classical cosmology to a dynamic, expanding universe. By deconstructing the historical myth of the "static universe" and introducing Georges Lemaître's pioneering mathematical deductions, we bridge the gap between intuitive physical analogies and the rigorous framework of General Relativity.

1 THE MYTH OF THE UNCHANGING COSMOS

1.1 The Concept

Prior to the early 1920s, the prevailing cosmological consensus—shared even by Albert Einstein—was that the universe was static, eternal, and geometrically invariant. Observational astronomy of the era was largely confined to the Milky Way, leading to the assumption that the night sky represented a fixed canopy of stars in a steady state.

Mathematically, Einstein sought to apply his newly formulated theory of General Relativity (1915) to the cosmos as a whole. The core of his theory is encapsulated in the Einstein Field Equations (EFE), which relate the geometry of spacetime to the energy and momentum within it:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \tag{1}$$

where $G_{\mu\nu}$ is the Einstein tensor describing spacetime curvature, G is Newton's gravitational constant, c is the speed of light, and $T_{\mu\nu}$ is the stress-energy tensor.

However, assuming a universe filled with ordinary, pressureless matter (dust), gravity is strictly attractive. A static solution requires the scale factor of the universe, $a(t)$, to be constant ($\dot{a} = 0$ and $\ddot{a} = 0$). To prevent the universe from collapsing under its own gravity, Einstein introduced a mathematically permissible integration constant, the cosmological constant Λ, modifying his equations to:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \tag{2}$$

Here, Λ provided a repulsive counter-force to balance gravitational attraction exactly.

1.2 The Takeaway

The notion of a dynamic universe with a definitive temporal origin (a "birth day") was initially viewed not merely as physically unintuitive, but as mathematically repulsive. Einstein's static universe, however, was later proven to be violently unstable—akin to a pencil balanced on its tip; the slightest perturbation would cause it to either expand infinitely or collapse entirely.

2 THE PRIEST WHO MOVED THE UNIVERSE

2.1 The Concept

The conceptual breakthrough arrived via Georges Lemaître (and independently, Alexander Friedmann). A physicist and Catholic priest, Lemaître scrutinized the EFE without the arbitrary constraint of a static universe. By assuming the universe is homogeneous and isotropic on large scales (the Cosmological Principle), the geometry of spacetime is described by the Friedmann-Lemaître-Robertson-Walker (FLRW) metric:

$$ds^2 = -c^2 dt^2 + a^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 (d\theta^2 + \sin^2\theta d\phi^2) \right] \tag{3}$$

where $k \in \{-1, 0, +1\}$ represents the spatial curvature, and $a(t)$ is the time-dependent scale factor.

Substituting the FLRW metric into the EFE yields the Friedmann equations. The first Friedmann equation governs the expansion rate:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3} \tag{4}$$

Lemaître realized that the universe is actively expanding ($\dot{a} > 0$). Furthermore, tracing the dynamical evolution backward in time ($t \to 0$) under standard thermodynamic assumptions implies that the scale factor $a(t) \to 0$. At this limit, the energy density $\rho \to \infty$ and spacetime curvature becomes singular.

Lemaître termed this initial, unimaginably dense, and hot quantum state the "Primeval Atom"—a singular quantum from which the fabric of spacetime itself decayed and expanded.

2.2 The Takeaway

The "Big Bang" was not initially proposed as an explosion in space, but rather as a mathematical inevitability derived from the dynamics of spacetime. It was a rigorous retrodiction of the expanding universe solution to General Relativity.

3 THE BALLOON ANALOGY (UNDERSTANDING EXPANSION)

3.1 The Concept

A common pedagogical hurdle is conceptualizing expansion without a topological "edge" or center. To resolve this, we map the mathematical expansion of the FLRW metric to the classic two-dimensional balloon analogy.

Consider galaxies at fixed *comoving coordinates* χ_1 and χ_2. The physical (proper) distance $D(t)$ between them at cosmic time t is given by:

$$D(t) = a(t)\Delta\chi \tag{5}$$

Taking the time derivative yields the recessional velocity v:

$$v = \dot{D}(t) = \dot{a}(t)\Delta\chi = \frac{\dot{a}(t)}{a(t)}\left[a(t)\Delta\chi\right] = H(t)D(t) \tag{6}$$

where $H(t) \equiv \dot{a}/a$ is the Hubble parameter. This is the theoretical derivation of Hubble's Law ($v = H_0 D$ today).

3.2 The Takeaway

Galaxies are not moving through a pre-existing Euclidean void like shrapnel from a localized explosion. Their peculiar velocities (local motion through space) are small. The observed recessional velocities are the result of *metric expansion*—the very fabric of space, quantified by $a(t)$, is stretching between gravitationally unbound systems. This mathematically permits apparent recessional velocities $v > c$ at high redshifts without violating Special Relativity.

4 THE SMOKING GUN: THE COSMIC MICROWAVE BACKGROUND

4.1 The Concept

If the universe evolved from Lemaître's Primeval Atom, early epochs were characterized by extreme temperatures and densities, dominated by a coupled thermal bath of photons and baryons. The energy density of radiation scales inversely with the fourth power of the scale factor ($\rho_r \propto a^{-4}$), while its temperature scales inversely with the scale factor:

$$T(t) \propto \frac{1}{a(t)} \tag{7}$$

In the early universe, the ambient thermal energy $k_B T$ greatly exceeded the binding energy of hydrogen (13.6 eV). The universe was a hot, opaque plasma where photons were continuously scattered by free electrons via Thomson scattering.

As the universe expanded, it cooled. At a redshift of $z \approx 1100$, the temperature dropped to $T \approx 3000$ K, allowing electrons and protons to form neutral hydrogen. This event is known as *Recombination*. Consequently, the mean free path of photons effectively became infinite. Photons decoupled from matter and have been streaming freely through space ever since.

Because the scale factor has increased by a factor of roughly 1100 since decoupling ($a_0/a_{\text{rec}} = 1+z \approx 1100$), the original 3000 K blackbody radiation has been cosmologically redshifted. The peak wavelength shifted into the microwave regime, yielding an observed background temperature today of:

$$T_0 = \frac{T_{\text{rec}}}{1+z} \approx 2.725 \text{ K} \tag{8}$$

4.2 The Takeaway

Discovered by Penzias and Wilson in 1964, the Cosmic Microwave Background (CMB) is the relic radiation from the moment of decoupling. It perfectly matches a theoretical blackbody spectrum as defined by Planck's Law. The CMB serves as the ultimate empirical validation of the hot Big Bang model, proving Lemaître's hypothesis of a dense, primordial origin.

THE QUANTUM ENGINE (TOWARDS SUPERKNOWLEDGE)

Having established that the macroscopic universe is expanding from a dense, hot primordial state, we must now address the nature of that initial state itself. Classical General Relativity breaks down at $t = 0$, leading to a singularity. To understand the true genesis of the cosmos, we must transition from classical macrocosmology to the quantum mechanics that governed the earliest epochs. We will find that the concept of "nothing" is physically meaningless; the vacuum is, in fact, the ultimate engine of reality.

5 THE QUANTUM BOILING POT (QUANTUM FOAM)

5.1 The Concept

In classical physics, a vacuum is defined as a space entirely devoid of matter and energy. However, through the lens of Quantum Field Theory (QFT), this classical "quiet ocean" is a fundamentally flawed concept.

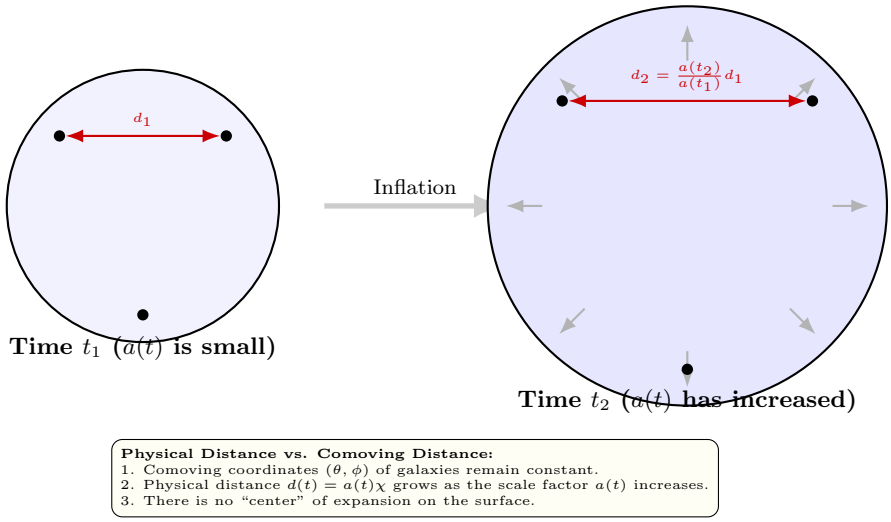

Figure 1: The Balloon Analogy: As the 2D surface of the balloon inflates (representing the scale factor $a(t)$ increasing), the coins (representing galaxies at fixed comoving coordinates) move further apart. The expansion occurs everywhere on the surface, with no definitive center.

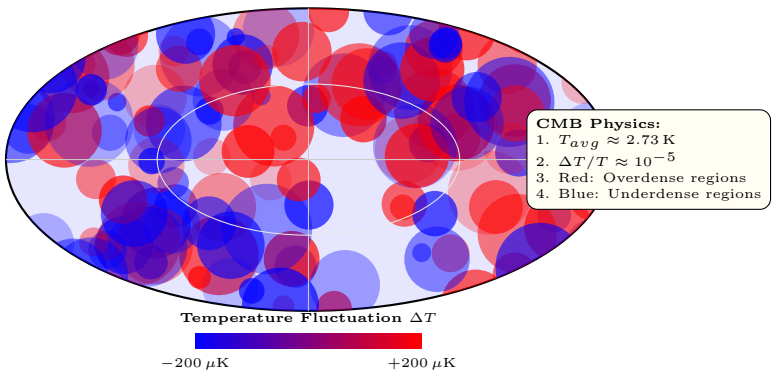

Figure 2: The Cosmic Microwave Background: The all-sky temperature map (e.g., from the Planck satellite) showing microscopic anisotropies ($\Delta T/T \sim 10^{-5}$) that served as the gravitational seeds for large-scale structure formation.

A quantum field can be modeled as an infinite set of coupled harmonic oscillators. The energy levels of a quantum harmonic oscillator are quantized as $E_n = (n + 1/2)\hbar\omega$. The lowest possible energy state, the vacuum state $|0\rangle$ where $n = 0$, is not zero. Instead, it possesses a non-zero zero-point energy:

$$E_0 = \frac{1}{2}\hbar\omega \tag{9}$$

Integrating this zero-point energy over all possible momentum modes up to a cutoff scale yields a massively divergent vacuum energy density.

Furthermore, Heisenberg's Uncertainty Principle allows for temporary violations of energy conservation, provided they occur over sufficiently short time scales:

$$\Delta E \Delta t \geq \frac{\hbar}{2} \tag{10}$$

This permits the spontaneous creation and annihilation of virtual particle-antiparticle pairs. At the Planck scale ($l_P \approx 1.6 \times 10^{-35}$ m), John Archibald Wheeler postulated that these energy fluctuations are so violent that the topology of spacetime itself tears and reconnects. This highly active, jittery state is known as *Quantum Foam*.

5.2 The Takeaway

The pre-Big Bang state was not a static emptiness. It was a highly active, fluctuating quantum field. The "vacuum" is a boiling pot of latent energy, setting the stage for a microscopic fluctuation to transition into a macroscopic reality.

6 COSMIC INFLATION: THE SPARK THAT LIT THE BANG

6.1 The Concept

How does a microscopic quantum jitter evolve into an observable universe containing hundreds of billions of galaxies? The mechanism is *Cosmic Inflation*, proposed by Alan Guth in 1980.

Inflation posits a brief, violent period of exponential expansion (10^{-36} to 10^{-32} seconds after the Planck epoch). This expansion is driven by a hypothesized scalar field, the *inflaton* (ϕ). The dynamics of the inflaton are governed by its potential energy density $V(\phi)$. During inflation, the scalar field rolls slowly down its potential, acting as an effective cosmological constant. The Friedmann equation during this epoch simplifies to:

$$H^2 \approx \frac{8\pi G}{3} V(\phi) \tag{11}$$

Assuming the "slow-roll" conditions ($\dot{\phi}^2 \ll V(\phi)$ and $\ddot{\phi} \ll 3H\dot{\phi}$), the Hubble parameter H is nearly constant. This yields an exponential solution for the scale factor:

$$a(t) \propto e^{Ht} \tag{12}$$

During this period, space expands by a factor of at least e^{60} (60 e-folds).

Crucially, the inflaton field is subject to quantum fluctuations $\delta\phi$. As the universe inflates exponentially, these microscopic quantum fluctuations are stretched beyond the Hubble horizon, freezing out as macroscopic, classical density perturbations $\frac{\delta\rho}{\rho}$.

6.2 The Takeaway

Inflation is the spark that lit the bang. It provides the mechanism to stretch the quantum boiling pot into a macroscopic arena. The galaxies, superclusters, and cosmic voids we observe today are not random; they are the gravitationally magnified fossils of pre-Big Bang quantum fluctuations.

7 THE ETERNAL BUBBLE BATH (MULTIVERSE THEORY)

7.1 The Concept

If inflation can occur via a scalar field rolling down a potential, we must account for the probabilistic nature of quantum mechanics governing that field. This leads to the paradigm of *Eternal Inflation*.

During the slow-roll phase, the classical motion of the field is driven down the potential ($\Delta\phi_{\text{class}} \approx \frac{\dot{\phi}}{H}\Delta t$). Simultaneously, the field experiences quantum fluctuations with an amplitude of roughly $\delta\phi \approx \frac{H}{2\pi}$ per Hubble time.

If the potential is sufficiently flat, there will be regions of space where the quantum "kick" dominates the classical rolling, pushing the field *up* the potential to a higher energy state:

$$\delta\phi > \Delta\phi_{\text{class}} \implies \frac{H}{2\pi} > \frac{\dot{\phi}}{H} \tag{13}$$

In these regions, inflation does not end; it accelerates. Because inflating space expands exponentially faster than non-inflating space, the total volume of the inflating background grows indefinitely.

Inflation only ends locally in localized "bubbles" where the scalar field successfully rolls to the true minimum of its potential (a process known as false vacuum decay, mathematically described by Coleman-De Luccia instantons). When the field reaches the minimum, its latent energy is dumped into the creation of standard matter and radiation—this is the process of *reheating*, which we classically call the Big Bang.

7.2 The Takeaway

Our Big Bang might not be the singular beginning of everything, but just *a* beginning. Our universe could simply be one expanding "bubble" (where inflation has gracefully exited) floating in a much larger, eternally expanding pre-existing quantum vacuum.

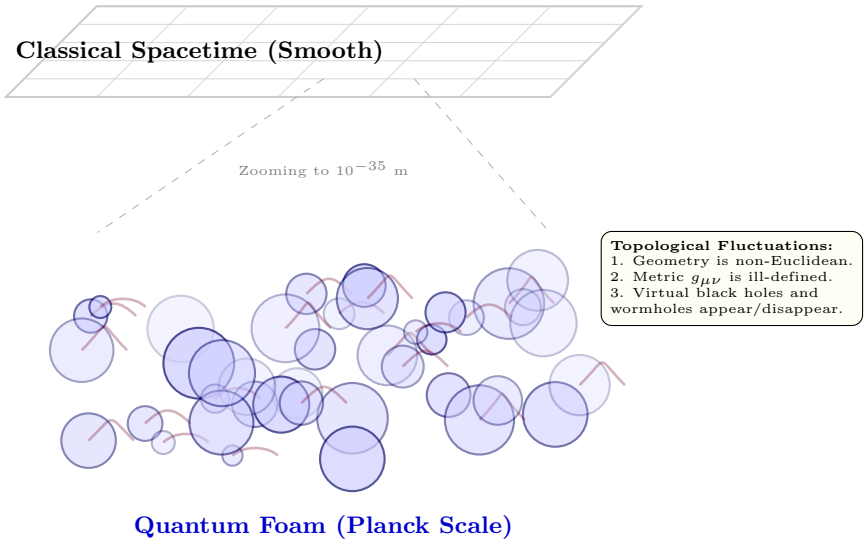

Figure 3: Conceptual representation of Quantum Foam. At the Planck scale, spacetime loses its smooth classical manifold structure (top), dominated instead by violent, probabilistic topological fluctuations (bottom) where bubbles and handles represent changing connectivity.

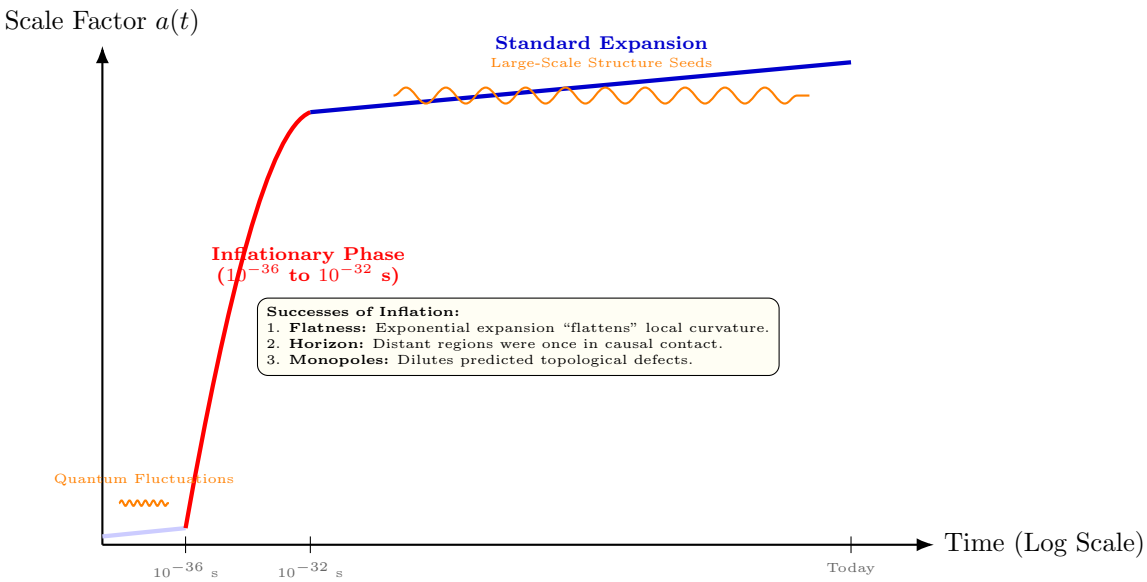

Figure 4: The timeline of Cosmic Inflation. A microscopic patch of the universe undergoes exponential expansion, flattening the local geometry and stretching quantum fluctuations to macroscopic scales. This phase is followed by the standard radiation- and matter-dominated expansion eras.

8 THE ULTIMATE FRONTIER: QUANTUM GRAVITY

8.1 The Concept

While inflation and QFT offer profound insights into the early universe, they are semi-classical approximations. We are treating quantum fields (matter and energy) on a classical, smooth background spacetime (General Relativity).

To truly map the pre-Big Bang vacuum at $t = 0$, we must formulate a complete theory of *Quantum Gravity*. At the Planck scale ($l_P = \sqrt{\frac{\hbar G}{c^3}}$), the curvature of spacetime is so extreme that General Relativity yields infinities, and standard QFT becomes non-renormalizable.

The preliminary mathematical step to unite these domains is the canonical quantization of gravity, leading to the Wheeler-DeWitt equation:

$$\hat{\mathcal{H}}\Psi[g_{\mu\nu}, \phi] = 0 \tag{14}$$

where $\hat{\mathcal{H}}$ is the Hamiltonian constraint operator and Ψ is the "wavefunction of the universe." Notice the absence of a time derivative; time itself disappears in this quantum gravitational framework, leading to the famous "Problem of Time." Current efforts to resolve this and provide a UV-complete theory rely on frameworks like String Theory or Loop Quantum Gravity (LQG).

8.2 The Takeaway

Quantum Gravity is the final boss of cosmology. Until we have a working, testable theory that successfully marries the physics of the incredibly massive with the incredibly small, the exact nature of the "Before Time" vacuum remains the greatest unresolved mystery in theoretical physics.

SUMMARY (FROM STATIC COSMOS TO QUANTUM ORIGINS)

The journey through this exposition traces the intellectual evolution of modern cosmology—from the classical misconception of a static universe to the dynamic, quantum-driven picture of cosmic genesis.

Early twentieth-century cosmology assumed an eternal and unchanging universe. Even Albert Einstein initially modified his field equations by introducing the cosmological constant Λ to artificially maintain a static cosmos. However, the solutions discovered by Friedmann and later developed by Georges Lemaître demonstrated that General Relativity naturally predicts a dynamical universe whose scale factor $a(t)$ evolves with time.

Lemaître's insight that the universe must have originated from an extremely dense and hot state—the "Primeval Atom"—laid the conceptual foundation for what would later become known as the Big Bang theory. Observational confirmation arrived through the discovery of the expansion of the universe and, most decisively, the detection of the Cosmic Microwave Background radiation. The CMB represents the fossil radiation of the early plasma epoch, providing compelling empirical validation of the hot Big Bang framework.

Understanding the expansion required a shift in intuition: galaxies are not flying through space from a central explosion; rather, space itself is expanding. The FLRW metric and the Friedmann equations quantitatively describe this metric expansion, leading naturally to Hubble's law.

Moving deeper into the earliest epochs, classical General Relativity alone becomes insufficient. Near the Planck scale, quantum effects dominate and spacetime itself becomes subject to fluctuations. Quantum Field Theory reveals that the vacuum is not empty but filled with zero-point energy and transient particle fluctuations. This dynamic vacuum structure, often described as quantum foam, provides the microscopic stage from which macroscopic cosmology may emerge.

Cosmic inflation offers the mechanism that bridges these scales. A brief period of exponential expansion driven by the inflaton field stretched microscopic quantum fluctuations into macroscopic density perturbations. These perturbations later evolved into the galaxies, clusters, and cosmic structures observed today.

In more speculative extensions, the theory of eternal inflation suggests that our observable universe may be only one bubble within a much larger multiverse, where inflation continues indefinitely in other regions of spacetime.

Despite these profound advances, the ultimate origin of the universe remains unresolved. At the Planck scale, where spacetime curvature becomes extreme, General Relativity and Quantum Mechanics must be unified within a complete theory of quantum gravity. Whether through String Theory, Loop Quantum Gravity, or another yet-unknown framework, this unification remains the central challenge of theoretical physics.

Thus, the Big Bang should not be viewed merely as an explosion in space, but as the beginning of cosmic expansion emerging from deep quantum processes. The quest to understand this transition—from quantum vacuum fluctuations to a structured universe—continues to push the frontiers of cosmology and fundamental physics.